A Practical activity report submitted

for

Engineering Design Project-II (UTA-024)

by

Name : Ishaan Setia
Roll no. : 1024030879
Subgroup : 2C63

Submitted to : Dr. Amit

THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

THAPAR INSTITUTE OF ENGINEERING AND TECHNOLOGY
PATIALA, PUNJAB

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Experiment: 1

Objective:

- To draw a schematic diagram of receiver to receive specified pulse width IR signals from gantries using CAD tool (Eagle).
- To design a printed circuit board layout of receiver circuit using CAD tool (Eagle).

Software Used: Eagle Software

Component Used:

Sr. No	Name of Components	Value	Specifications	Quantity
1.	Resistor	120k Ω	Carbon Resistor with 5% Tolerance	1x
2.	Resistor	100k Ω	Carbon Resistor with 5% Tolerance	1x
3.	Resistor	22k Ω	Carbon Resistor with 5% Tolerance	1x
4.	Resistor	1k	Carbon Resistor with 5% Tolerance	1x
5.	SchottkyDiode (BPW41N)	N/A	Silicon PIN Schottky diode,	1x
6.	Male Header	3-pin	PCB Header	1x
7.	Operational Amplifier (LM311N)	N/A	DIP-8 Package	1x

Theory:

1. Resistor:

Aresistor is an electronic component designed to oppose or limit the flow of electric current ina circuit. It works by providing resistance, measured in ohms (Ω), which controls how much current can pass through. The relationship between voltage, current, and resistance is described by Ohm's Law: V=I×RV = I \times RV=I×R, where VVV is voltage, III is current, and RRR is resistance. Resistors are essential for controlling current levels, dividing voltages, protecting sensitive components from excessive current, and working with other components like capacitors to set timing in circuits. They come in different types, such as fixed resistors with constant resistance, variable resistors (potentiometers) with adjustable resistance, and special types like light-dependent resistors (LDRs) or thermistors, which change resistance based on environmental conditions. In circuit diagrams, resistors are represented either by a zig-zag line (US style) or a rectangle (international style)

Fig. 1.1 various types of resistors

2. Schottky Diode:

A Schottky diode is a special type of semiconductor diode that uses a metal– semiconductor junction instead of the traditional p–n junction found in standard diodes. This design gives it a much lower forward voltage drop, typically between 0.15 V and 0.45 V, compared to around 0.7 V for a regular silicon diode. Because of this low voltage drop, Schottky diodes can conduct current more efficiently and generate less heat. Another key advantage is their very fast switching speed, which makes them ideal for highfrequency applications such as radio frequency (RF) circuits, power rectifiers in switching power supplies, and clamping circuits. However, they generally have a lower reverse voltage rating and higher reverse leakage current compared to regular diodes, meaning they are less suited for high-voltage blocking applications. In circuit diagrams, they are represented like a standard diode symbol but with an "S"-shaped bend or thickened line on the cathode side to indicate their special junction

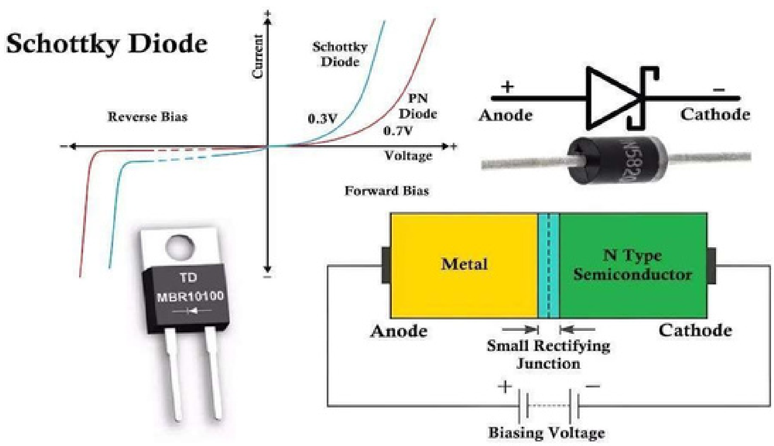


Fig. 1.2 Schottky diode

3. Led 3mm: (3)

A3 mm LED is a small light-emitting diode with a lens diameter of 3 millimeters, commonly used as an indicator light in electronic circuits. Like all LEDs, it is a semiconductor device that emits light when an electric current passes through it in the forward direction. The light color depends on the semiconductor material, with common options including red, green, blue, yellow, and white. A 3 mm LED is compact, making it suitable for space-limited projects such as control panels, small electronic gadgets, model lighting, and hobby circuits. It typically has two leads: the longer lead (anode) is the positive side, and the shorter lead (cathode) is the negative side. LEDs require a series resistor to limit current, since applying too much current can burn them out. The forward voltage of a 3 mm LED usually ranges from about 1.8 V for red to around 3.2 V for blue.

Fig. 1.3 Various types of sub miniature standardLED

4. Male Header: (4)

A male header is an electrical connector consisting of a row (or multiple rows) of metal pins mounted in a plastic base, designed to plug into a matching female header or socket. The pins are usually spaced at standard intervals, most commonly 2.54 mm (0.1 inch), making them compatible with breadboards, perfboards, and many PCB layouts. Male headers are widely used in electronics for creating modular connections between circuit boards, attaching sensors or modules to microcontrollers, and enabling easy disconnection for repairs or upgrades. They can come in straight or right-angle orientations, single or multiple rows, and in various pin counts depending on the application. The exposed metal pins allow for easy soldering or insertion into connectors, but also require careful handling to avoid bending or short-circuits.

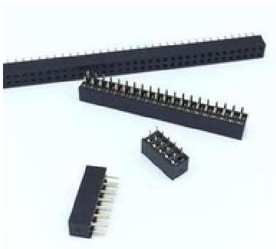


Fig. 1.4 Male PCB Header

5. Operational Amplifie(LM311N):

The LM311N operational amplifier is a singular comparator device that facilitates the comparison of dual input voltages, subsequently generating an output predicated on their differential magnitude. Exhibiting a response time of 200 nanoseconds and a supply current of 1.5 milliamperes, this device operates with elevated speed and diminished power consumption. Furthermore, the LM311N boasts a broad supply voltage range of ±5 to ±15volts, accompanied by an input impedance of 10^6 ohms and an output impedance of 50 ohms. Its hysteresis is readily adjustable via external resistors, and its output is compatible with TTL/CMOS logic. This device finds application in a myriad of electronic circuits, including voltage comparators, level detectors, pulse generators, oscillators, analog-to-digital converters, and digital-to-analog converters, underscoring its versatility and efficacy in various technological contexts.

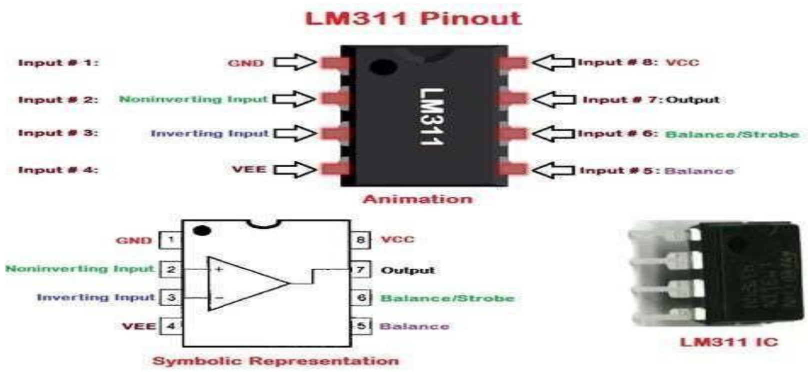


Fig. 1.5 LM311N

Schematic diagram:

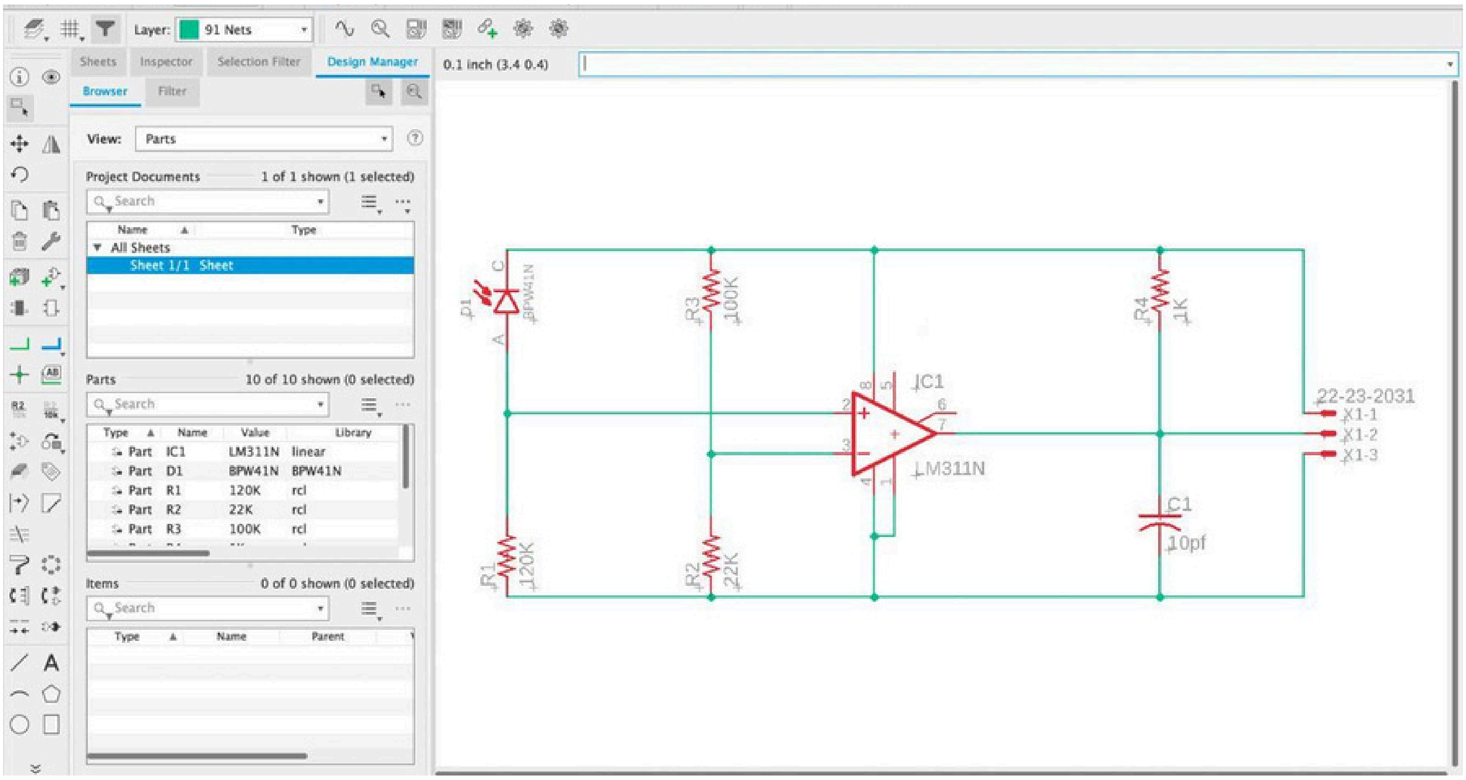


Fig. 1.6 Schematic diagram of Receiver

Printed Circuit Board layout:

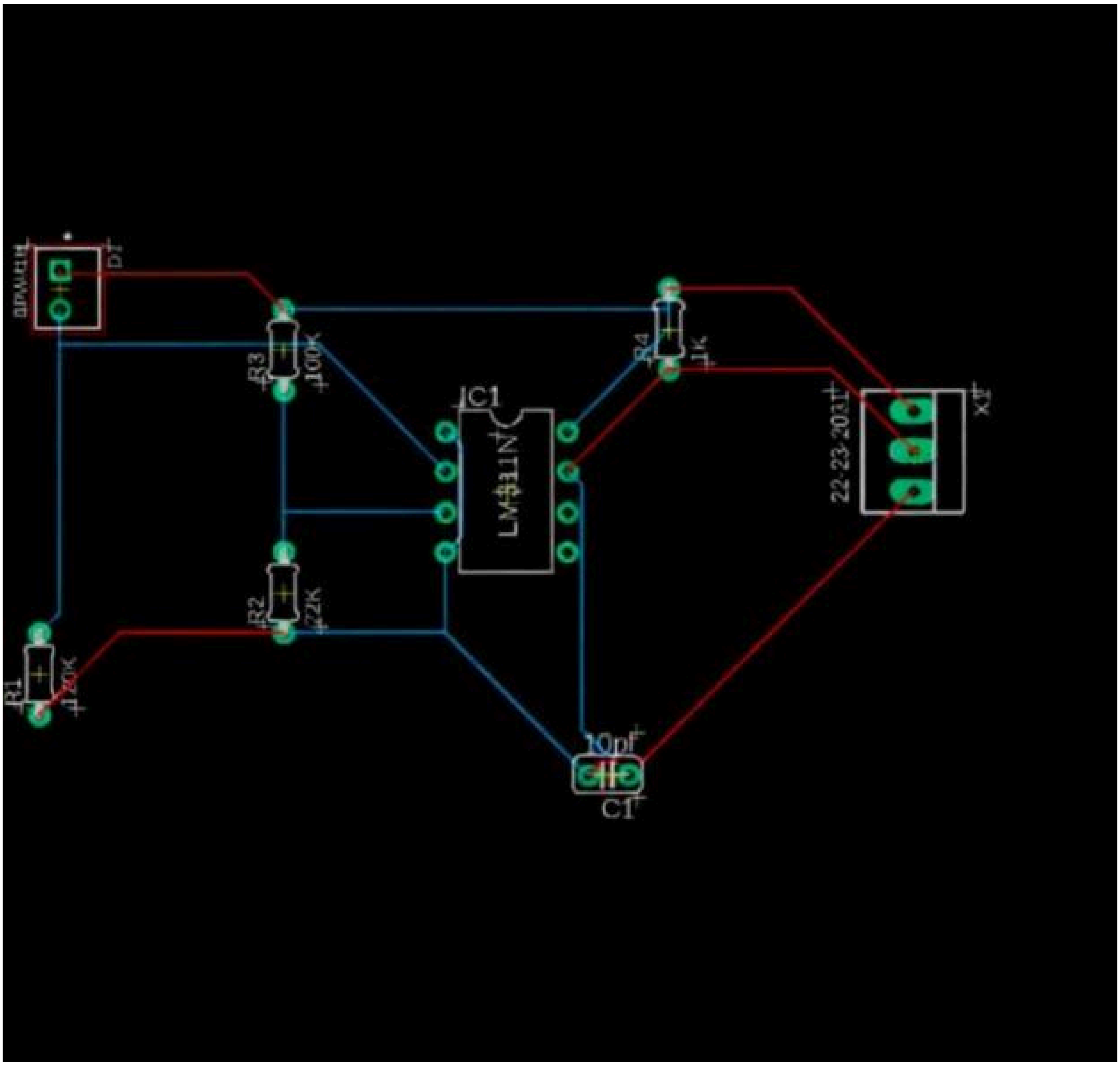
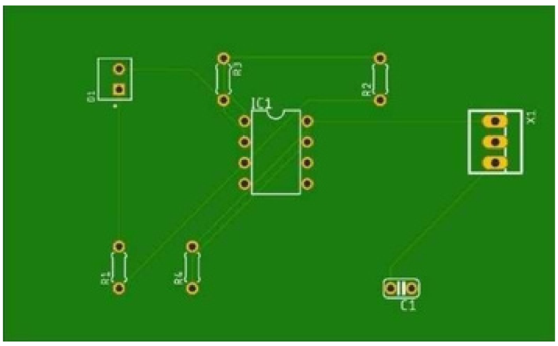


Fig. 1.7 PCB layout of Receiver circuit



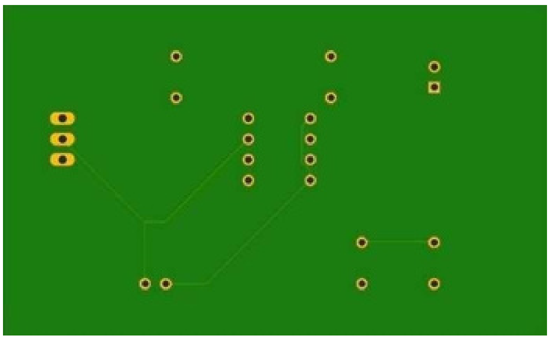


Fig. 1.8 Top and Bottomside PCB view

Procedure:

1. Create a project by going to file option > new > project.
2. Give name to the newly created project.
3. To add a schematic to a project folder, right-click the folder, hover over "New" and select "Schematic".
4. Add parts to the schematic using ADD tool and wire up the components using NET tool.
5. Provide the name and the values.
6. Switch from the schematic editor to the related board by simply clicking the switch to board command.
7. Now arrange all the components.
8. Select the bottom layer and select manual ROUTE Tool and do the outlining.
9. Now select auto routing then select continue > start > end job >manufacturing >end job.
10. Save the top and bottom view.

Discussion:

In this experiment, a receiver circuit was developed to detect specific pulse width infrared (IR) signals from gantries. The design process utilized Eagle for creating and simulating the schematic diagram and PCB layout.

After designing and simulating the circuit, the PCB was fabricated, populated, and tested. The results showed the circuit effectively detected the desired IR signals. Eagle's use streamlined the design and testing processes, highlighting its value in electronic design. Future work may involve improving the circuit's sensitivity and integrating it into a more comprehensive IR detection system, as well as exploring other CAD tools for further optimization.

Reference:

[1] https://www.electricaltechnology.org/2015/01/resistor-types-resistors-fixed-variable-linear-non-linear.html

[2] https://pin.it/69NT73k62

[3] https://upload.wikimedia.org/wikipedia/commons/7/7f/SS14_1A_DO-214_Schottky_diode.jpg

[4] https://i.pinimg.com/564x/81/d5/32/81d532c2afb7546a7b6281c207cffd26.jpg

[5] https://i.pinimg.com/564x/aa/cd/ca/aacdca6efe10f52beb0240d0fdbe7cea.jpg

REFERENCE FOR TEXT:

(1) https://www.allaboutcircuits.com/textbook/direct-current/chpt2/resistance/ (2) https://www.vishay.com/docs/81522/bpw41n.pdf (3) https://www.electronics-tutorials.ws/io/io_4.html (4) https://www.molex.com/en-us/products/connectors/wire-to-boardconnectors (5) https://www.ti.com/lit/ds/symlink/lm311.pdf

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Experiment: 2

Objective:

- To draw a schematic diagram of IR sensor module circuit (required to move Buggy module on a predefined the path) using CAD tool (Eagle).
- To design a printed circuit board layout of IR sensor module circuit using CAD tool (Eagle).

Software Used: Eagle Software

Component Used:

Sr. No	Name of Components	Value	Specifications	Quantity
1.	Resistor	220 Ω	Carbon Resistor with 5% Tolerance	4x
2.	Resistor	10k Ω	Carbon Resistor with 5% Tolerance	2x
3.	Potentiometer	10k Ω	Linear Potentiometer (Preset / Variable Resistor)	2x
4.	led3mm	5V	Forward Voltage ≈ 2 V, 20 mA	2x
5.	IR Transmitter (SFH482)	N/A	Infrared LED, Wavelength ≈ 940 nm	2x
6.	IR Receiver (BPX65)	N/A	Silicon PIN Photodiode, IR Sensitive	2x
7.	Operational Amplifier LM358P	N/A	Dual Op-AmpIC, DIP-8 Package	1x

Theory:

1.IRTransmitter (BPX65):

Aninfrared (IR) transmitter is an electronic device that emits infrared light signals, typically inthewavelength range of 850 nm to 950 nm, which are invisible to the human eye but can bedetected by IR receivers. It usually consists of an infrared light emitting diode (IR LED) thatconverts electrical signals into modulated light pulses. These pulses carry information, suchas commands for TVs, air conditioners, or other remote-controlled devices. The modulation, often around 38 kHz, helps the receiver distinguish the signal from background infrared radiation (like sunlight). IR transmitters are widely used in remote controls, wireless data transfer systems, obstacle detection circuits, and short-range communication between devices. They are compact, energy efficient, and inexpensive, but require a direct line of sight to the receiver for reliable communication. In circuit diagrams, an IR transmitter is represented similarly to an LED symbol, with arrows pointing outward to indicate emitted light

Fig. 2.1 Various types of IR

2. IR Receiver (SFH482):

An infrared (IR) receiver is an electronic component designed to detect infrared light signals, usually in the wavelength range of 850 nm to 950 nm, and convert them into electrical signals that can be processed by a circuit. It is often used in conjunction with an IR transmitter for wireless communication, such as in TV remotes, air conditioner controls, and other remote-operated devices. Most IR receivers include a photodiode or phototransistor to sense the IR light, along with built-in signal processing circuitry that filters and demodulates the incoming signal, typically modulated at around 38 kHz to avoid interference from ambient light sources. The output is usually a digital signal that changes state when the correct IR pulses are detected. IR receivers are compact, inexpensive, and reliable for short-range, line-of-sight communication. In circuit diagrams, they are represented as a photodiode or a small sensor package with arrows pointing inward, indicating incoming light

Fig.2.2 Various types of IR

3. Potentiometer:

Apotentiometer,oftenreferredtoas a "pot," is a three-terminal variable resistor used to adjusttheflowofelectriccurrentby varying resistance. It works by moving a sliding contact(wiper)acrossauniformresistance element, which adjusts the voltage output based on thewiper'sposition.Potentiometers are commonly used as adjustable voltage dividers in applicationslikevolumecontrols on audio equipment, speed control of fans, and as positiontransducersinjoysticks.They come in two main types: rotary, which adjusts by rotating,andlinear,whichadjustsby sliding linearly.

Fig. 2.3 Potentiometer

4. Operational Amplifier (LM358P):

The LM358P is a dual operational amplifier integrated circuit known for its stability and cost-effectiveness. It features two independent high-gain operational amplifiers, which can operate from a single power supply over a wide range of voltages. This makes it versatile for various applications, including signal conditioning, filtering, and amplification. The LM358P is designed with internal frequency compensation and low power consumption, making it suitable for battery- operated devices. Its wide range of applications and reliable performance make it a popular choice in both commercial and industrial projects.

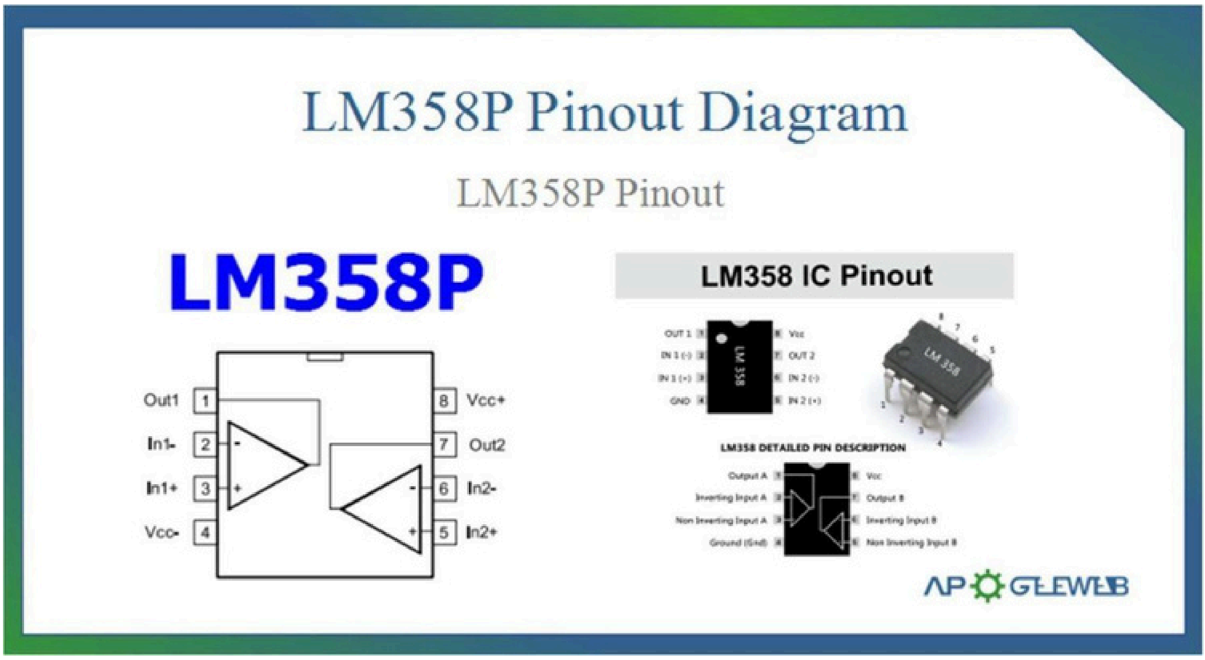


Fig. 2.4 LM358P

Schematicdiagram:

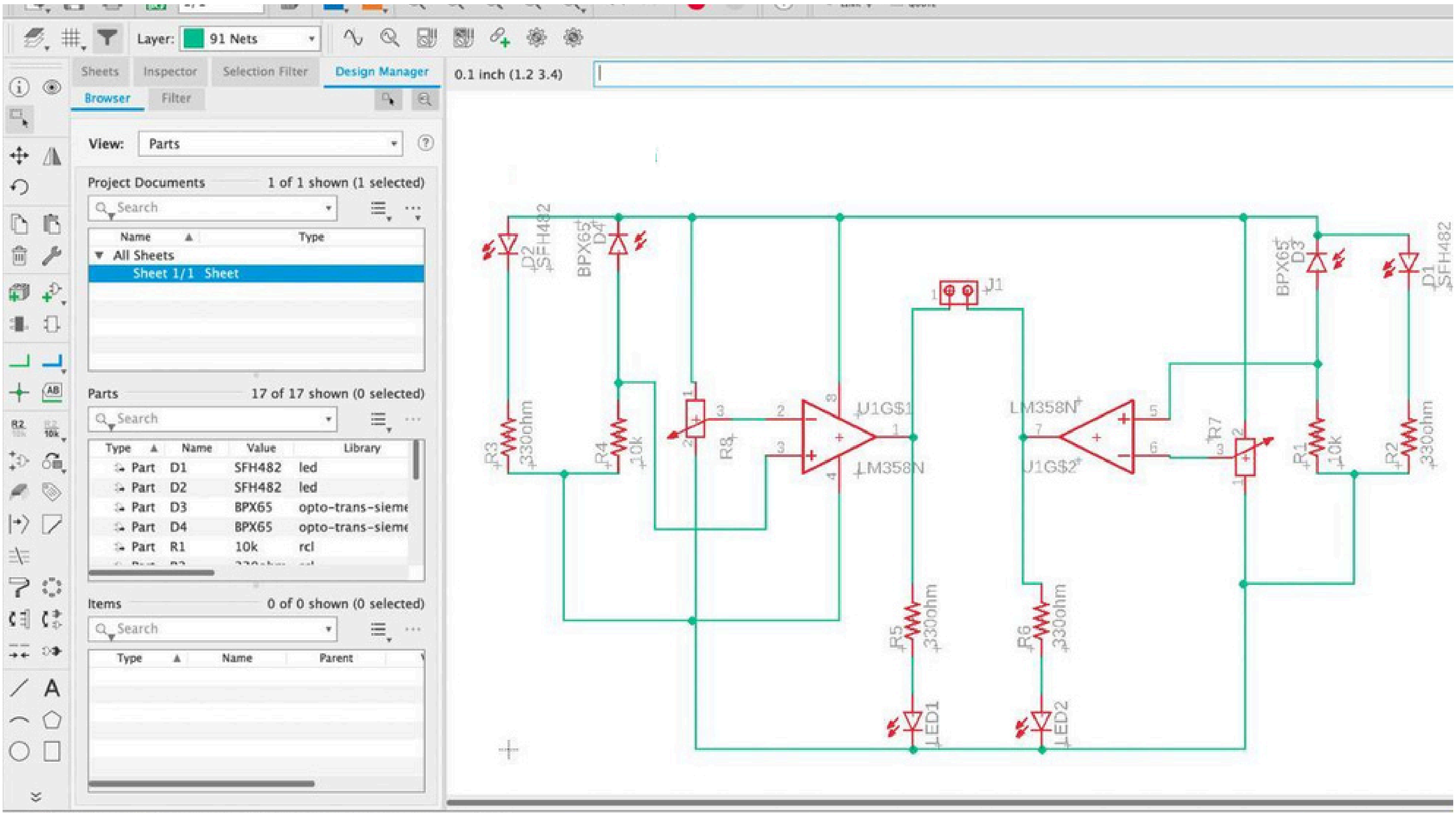


Fig. 2.5Schematic diagram of IR circuit

Printed Circuit Board layout:

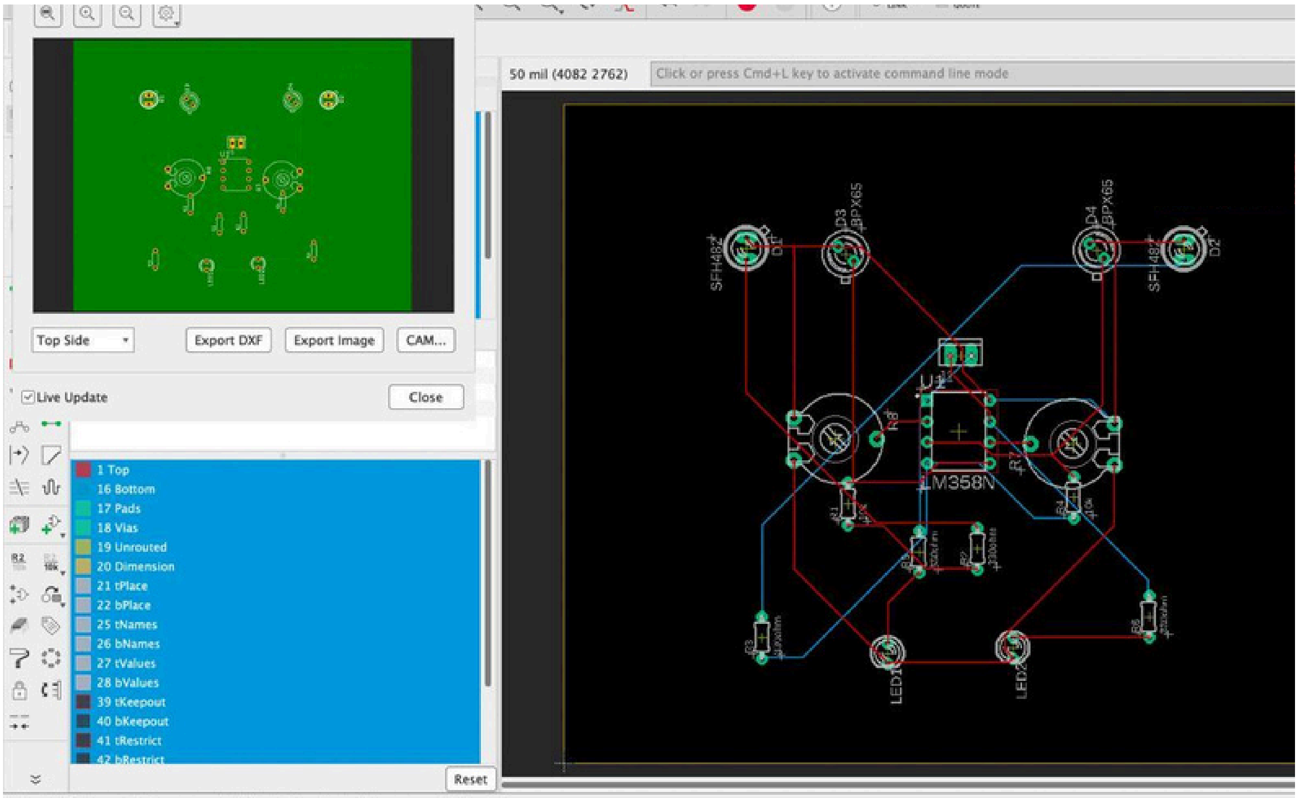


Fig. 2.6 PCB layout of IR circuit

Fig. 2.7 PCB layout of IR circuit (Top)

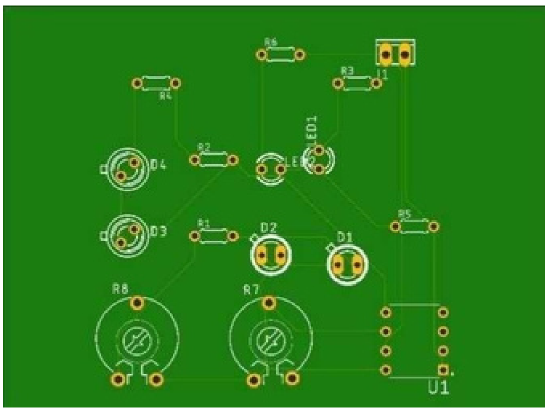
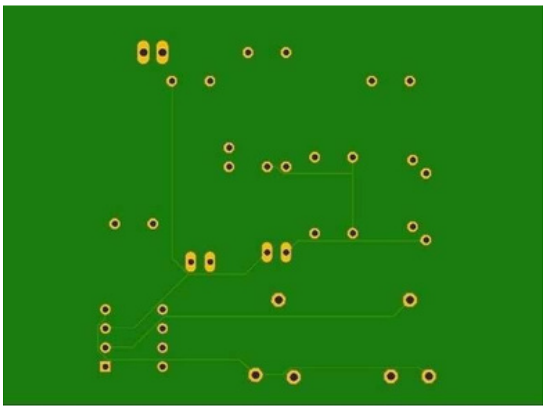


Fig. 2.8 PCB layout of IR circuit (Bottom)



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5. Provide the name and the values.
6. Switch from the schematic editor to the related board by simply clicking the switch to board command.
7. Now arrange all the components.
8. Select the bottom layer and select manual ROUTE Tool and do the outlining.

Discussion:

This experiment involves two primary tasks:

To create a detailed circuit diagram illustrating the interconnection of components, such as the IR sensor, resistors, capacitors, and any necessary microcontroller or interface circuitry.

The schematic should accurately represent the electrical connections and functionality required for the IR sensor to detect objects and guide the Buggy module's movement and designing a physical arrangement of components on a printed circuit board (PCB).

This layout should ensure proper component placement, trace routing, and grounding to minimize electrical interference and optimize the circuit's performance. The PCB layout should also consider factors like size, cost, and manufacturability.

REFERENCE FOR TEXT:

(1) https://www.osram.com/ecat/SFH%20482/com/en/class_pim_web_catalog_103489/
(2) https://www.vishay.com/docs/81521/bpx65.pdf
(3) https://www.electronics-tutorials.ws/resistor/potentiometer.html (4) https://www.ti.com/lit/ds/symlink/lm358.pdf

Reference:

[1] https://www.electricity-magnetism.org/infrared-transmitters/

[2] https://www.electronicshub.org/ir-transmitter-receiver-circuits/

[3] https://en.wikipedia.org/wiki/Potentiometer

[4] https://how2electronics.com/lm358-dual-op-amp-features-pins-working-applications/

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Experiment: 3

Objective:

- To draw a schematic diagram of pulse width modulation (PWM) based transmitter for generating specified pulse width waveforms for gantries placed at different locations on the path using CAD tool (Eagle).
- To design a printed circuit board layout of pulse width modulation (PWM) based transmitter using CAD tool (Eagle). Software Used: Eagle Software

Component Used:

Sr. No	Name of Components	Value	Specifications	Quantity
1.	Resistor	220 Ω	Carbon Resistor with 5% Tolerance	1x
2.	Capacitor	1000nF	Electrolytic Capacitor	1x
3.	Capacitor	10nF	Ceramic Capacitor	1x
4.	DCJ0202	NA	DC Power Jack	1x
5.	led3mm	5V	Dome Lamp	1x
6.	IC 78L05Z	5V	Positive Voltage Regulator	1x
7.	22-23-2031	NA	PCB Header	1x
8.	ATTINY85	NA	8-bitAVR Microcontroller, DIP-8 Package	1x

Theory:

1. Capacitor:

Acapacitor is an electronic componentthatstoreselectrical energy in an electric field and releases it when needed. It consistsoftwoconductiveplates separated by an insulating material called a dielectric. When a voltageisappliedacrosstheplates, electric charges build up on them, creating the stored energy. The amountofchargeacapacitor can store is measured in farads (F), with most practical capacitors beinginmicrofarads(μF), nanofarads (nF), or picofarads (pF). Capacitors are used for various purposes,suchassmoothing voltage in power supplies, filtering signals, coupling or decoupling ACandDCcomponents, and timing applications in combination with resistors. In circuit diagrams,acapacitorisrepresented by two parallel lines (for nonpolarized types) or one straightandonecurvedline (for polarized types like electrolytic capacitors).

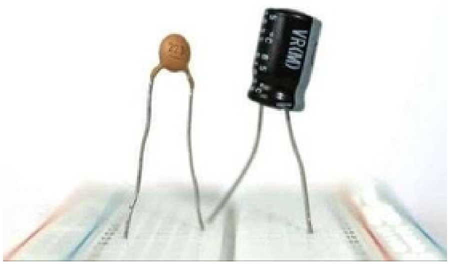


Fig. 3.1 Various types of capacitors

2. IC 78L05Z:

This IC is used to power low-current electronic circuits that require a stable 5V supply, protecting them from voltage fluctuations. It has built-in features like thermal shutdown and short-circuit protection, making it reliable and safe to use. The input voltage should typically be between 7V and

20V for proper regulation, and small capacitors are often connected at the input and output to improve stability and reduce noise

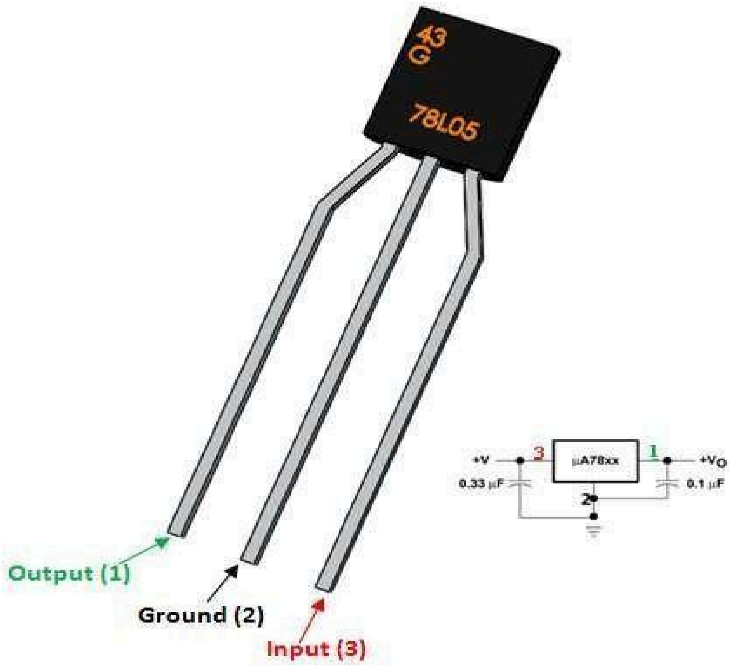


Fig.3.2Voltage regulator 78L05Z

3. IC ATTTINY85:

TheATtiny85isasmall,low-power8-bitmicrocontroller from Microchip’s (formerly Atmel’s) AVR family.Ithas8KBofflashmemory,512bytes of SRAM, and 512 bytes of EEPROM, making it suitableforcompactprojectsthatrequirebasic processing capabilities. Despite its small size, it offers usefulfeaturessuchas6I/Opins,ADC(Analog-to-Digital Converter), PWM outputs, timers, and supportforserialcommunicationprotocols like I²C and SPI.The ATtiny85 is popular for DIY electronics,wearables,smallautomationprojects, and battery powered devices due to its low power consumptionandabilitytorunatclockspeeds up to 20 MHz (with an external oscillator) or 8 MHz (internal).Itiscommonlyavailableinan8-pin DIP or SOIC package, making it easy to use on breadboardsorsolderintosmallcircuits.Developers often program it using the Arduino IDE with an external programmeroranotherArduinoboard as an ISP (In-System Programmer)

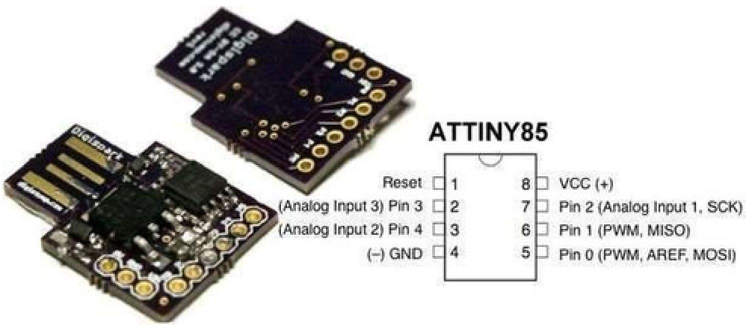


Fig. 3.3 IC ATTINY85

4. IC DCJ0202:

The DCJ0202 is a type of DC power jack used to provide a reliable connection between an external power supply and an electronic device. It is designed to receive power from a compatible DC plug, typically with an outer diameter of around 2.00 mm, and transfer it to the internal circuitry of the device. DC power jacks like the DCJ0202 are commonly used in small electronic equipment, development boards, and consumer devices where a stable, detachable power connection is needed. The jack usually consists of conductive contacts enclosed in an insulating housing, ensuring both electrical conductivity and mechanical durability. Its design allows easy plugging and unplugging while maintaining a secure fit to prevent accidental disconnection. Depending on the application, it can be mounted on a printed circuit board (PCB) using through-hole or surface-mount techniques

Fig. 3.4 IC DCJ0202

5. 22-23-2031(MTA02-100):

The Molex 22-23-2031 (MTA02-100) is a 3-pin male header connector from the KK 254 series, designed for reliable wire-to-board connections in electronic circuits. It features a standard 2.54 mm (0.1 inch) pitch, making it widely compatible with breadboards, PCBs, and other prototyping platforms. The connector uses a through-hole mounting style for secure soldering onto the board and includes a friction-lock mechanism that prevents accidental disconnection during operation. Constructed with a nylon housing and tin-plated brass contacts, it offers good electrical conductivity and mechanical durability. Rated for up to 4 A and 500 V, the 22-23-2031 is suitable for both signal transmission and low-power applications. Its operating temperature range of –40°C to +80°C, along with its UL 94 V-0 flammability rating, ensures safe and stable performance in various environments. This connector is commonly used in embedded systems, control panels, and other electronic assemblies requiring a compact, secure, and easy-to-use interface

Fig. 3.5 Series female wire connector

Schematic diagram:

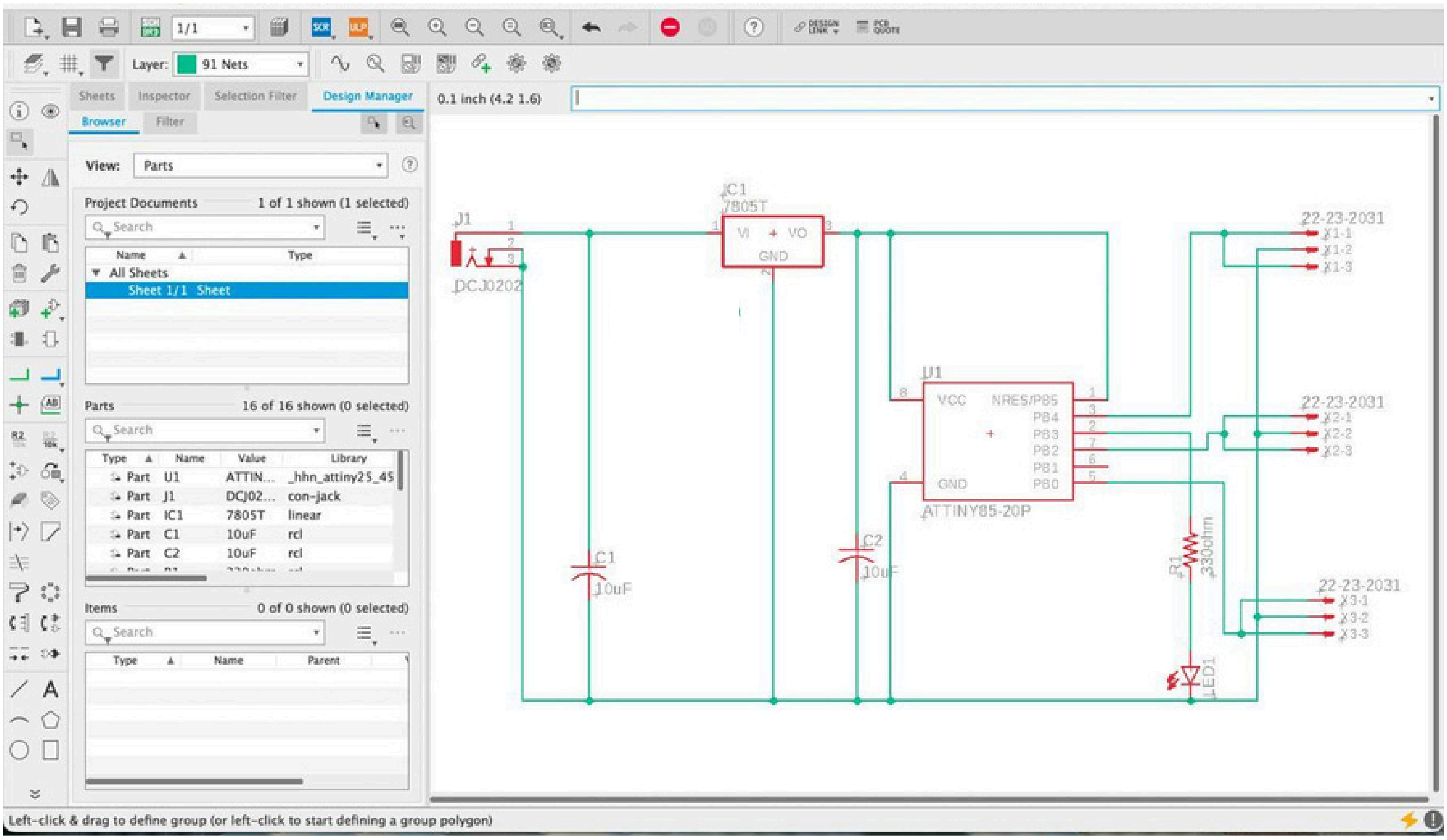


Fig. 3.6 Schematic diagram of Transmitter circuit

Printed CircuitBoard layout:

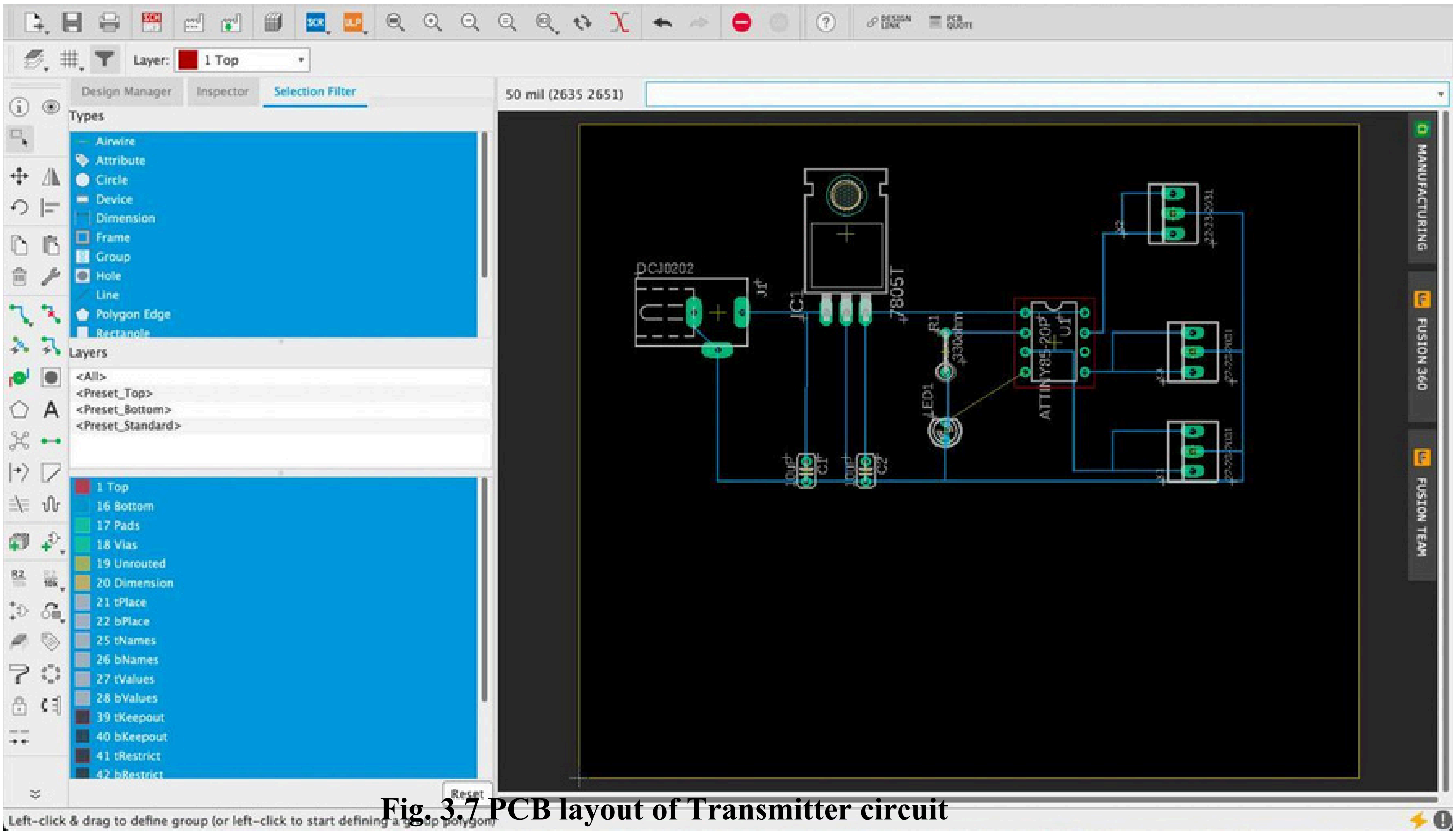
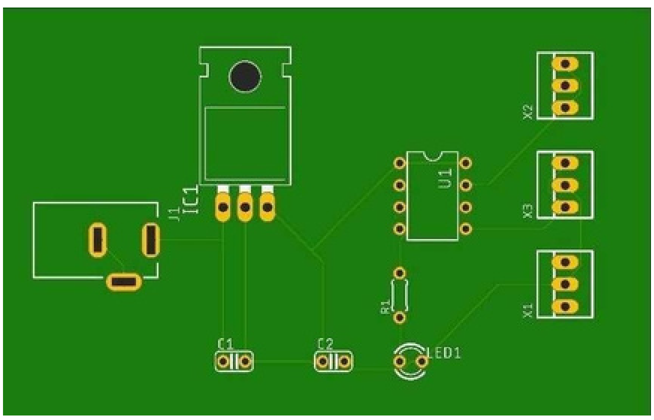


Fig. 3.7 PCB layout of Transmitter circuit



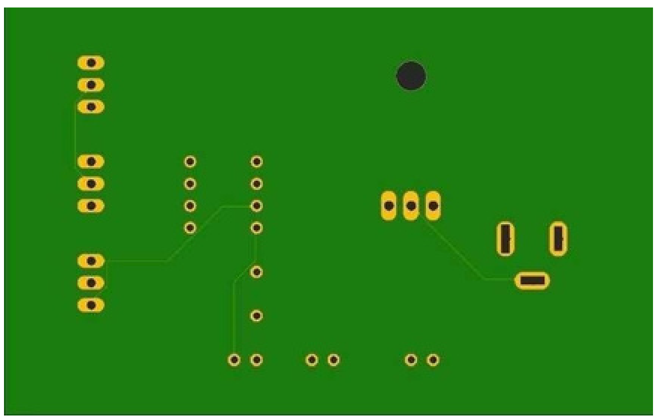


Fig. 3.8 PCB layout of Transmitter circuit (Top) **Fig. 3.9 PCB layout of Transmitter circuit (Bottom)**

Procedure:

1. Create a project by going to file option > new > project.
2. Give name to the newly created project.
3. To add a schematic to a project folder, right-click the folder, hover over "New" and select "Schematic".
4. Add parts to the schematic using ADD tool and wire up the components using NET tool.
5. Provide the name and the values.
6. Switch from the schematic editor to the related board by simply clicking the switch to board command.
7. Now arrange all the components.
8. Select the bottom layer and select manual ROUTE Tool and do the outlining.
9. Now select auto routing then select continue > start > end job >manufacturing >end job.
10. Save the top and bottom view.

Discussion:

This project involves the design and implementation of a PWM-based transmitter circuit that can produce precisely timed pulse width waveforms. These waveforms will serve as control signals for gantries, enabling their synchronized movement along the designated path. The use of Eagle CAD for both schematic design and PCB layout will ensure a systematic and efficient approach to the project. The schematic design will define the interconnection of components, such as the microcontroller, PWM generator, and output drivers. The PCB layout will determine the physical arrangement of these components on the printed circuit board, ensuring optimal signal integrity and manufacturability. By successfully completing this project, we will have a functional PWM-based transmitter capable

of generating the required pulse width waveforms to control gantries accurately and reliably. This system can be integrated into various applications, such as automated guided vehicles (AGVs) or industrial automation systems, where precise control of moving elements is essential.

Reference:

[1] https://en.wikipedia.org/wiki/Applications_of_capacitors

[2] https://www.digikey.com/en/products/detail/umw/78L05/17636890

[3] https://en.wikipedia.org/wiki/ATtiny_microcontroller_comparison_chart

[4] https://www.mouser.com/ProductDetail/Amphenol-CONEC/33DCJ-0202-A?qs=B9TV4DWH7gV7tdoDe%2F%252BtfA%3D%3D

[5]https://www.google.com/url?sa=i&url=https%3A%2F%2Fwww.automotiveconnectors.com%2F4-way-te-connectivity-mta-100-male-blue.html&psig=AOvVaw1pdXIAt85zfofAH1oQyK4G&ust=1724834583873000&source=images&cd=vfe&opi=89978449&ved=0CBcQjhxqFwoTCNiP1NLjlIgDFQAAAAAdAAAAABAE

REFERENCE FOR TEXT:

(1) https://www.electronics-tutorials.ws/capacitor/cap_1.html
(2) https://www.st.com/resource/en/datasheet/l78l05.pdf
(3) https://ww1.microchip.com/downloads/en/DeviceDoc/Atmel-2586-AVR-8-bitMicrocontrollerATtiny25-ATtiny45-ATtiny85_Datasheet.pdf
(4) https://www.cui.com/product/resource/dc-power-connectors.pdf
(5) https://www.molex.com/molex/products/part-detail/pcb_headers/0022232031

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